Sparse quadratic coding and shorter Pell finalizers for ZFC consistency

Explicit equations in nine positive or ten integer unknowns
and lower-degree eleven- and twelve-positive companions

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Abstract

We improve the explicit consistency equations in ProveIt packages 38–45 by using only the active quadratic source support, overlapping target digits, and an interior padding digit. The largest positional weight falls from 799 to 174. A new conditional-separation norm theorem, a second-order polynomial Pell majorant, and an exact CRT witness reparameterization substantially simplify the nine-variable finalizer. The recommended compromise uses nine positive unknowns, degree 10 486 092, and 603 live binary operations; a 655-operation companion lowers the degree to 10 127 484. A ten-unrestricted-integer companion has degree 33 202 776 in 689 operations; shared homogenization also gives degree 50 822 936 in only 582 operations. An independent lean-carrier construction gives eleven positive unknowns at degree 954 in 113 operations and twelve at degree 897 in 108 operations. Each lack-of-solutions assertion is provably equivalent in ZFC by the mathematical arguments below to the inherited syntactic Con(ZFC) predicate. We give the new proofs, literal polynomial registers, complete fixed coefficient payloads, finite positional certificates, and isolated reproducibility checks. We do not claim global minimality, a consistency decision, a new unknown-count theorem, or a newly checked end-to-end Lean, Rocq, or object-level ZFC derivation.

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1 Results and the meaning of simplest

The domain is part of an equation's specification. Throughout $\mathbb{N} = \{0, 1, \ldots\}$ and $\mathbb{Z}_{>0} = \{1, 2, \ldots\}$. An equation over $\mathbb{Z}_{>0}^{9}$ is not silently an equation over $\mathbb{Z}^{9}$.

Theorem 1.1 (Consistency presentations)**.** *Let the fixed integer polynomials in this report be instantiated by the complete coherent tuple in* `coefficients/COEFFICIENT_BINDING.json`*. Then, with the inherited compiler and arithmetic dependencies identified below, ZFC proves each of*

$$\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{9}\ F_9(\mathbf{x}) = 0 \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}),$$
$$\neg\exists \mathbf{x} \in \mathbb{Z}^{10}\ F_{10}(\mathbf{x}) = 0 \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}),$$
$$\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{11}\ P_{11}(\mathbf{x}) = 0 \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}),$$
$$\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{12}\ P_{12}(\mathbf{x}) = 0 \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

A separate coordinate translation gives the supplied nine-nonnegative form.

This is an equivalence theorem, not a proof of either truth value. The consistency predicate means the absence of a finite syntactic ZFC refutation under the pinned K215 calibration. No unconditional independence claim is inferred merely from consistency.

Domain	Presentation	Exact degree	Binary operations
$\mathbb{Z}_{>0}^{9}$	Prior package 45	129875980	788
$\mathbb{Z}_{>0}^{9}$	New interior regular	15502748	548
$\mathbb{Z}_{>0}^{9}$	New shifted packed, recommended	10486092	603
$\mathbb{Z}_{>0}^{9}$	New unshifted packed	10127484	655
$\mathbb{Z}^{10}$	Upstream package 40	237520456	782
$\mathbb{Z}^{10}$	Upstream package 38	449607016	773
$\mathbb{Z}^{10}$	New interior regular shared	50822936	582
$\mathbb{Z}^{10}$	New shifted packed shared, recommended	34378296	637
$\mathbb{Z}^{10}$	New unshifted packed	33202776	689
$\mathbb{Z}_{>0}^{11}$	Package 38, optional degree frontier	1140	129
$\mathbb{Z}_{>0}^{11}$	Package 38, main cost frontier	1154	123
$\mathbb{Z}_{>0}^{11}$	New lean Y absorption	954	113
$\mathbb{Z}_{>0}^{12}$	Package 44, full carrier	901	112
$\mathbb{Z}_{>0}^{12}$	Package 44, short carrier	905	110
$\mathbb{Z}_{>0}^{12}$	New lean carrier	897	108

The recommended nine-variable compromise reduces the prior degree by 91.93% and cost by 23.48%. The recommended ten-integer compromise reduces package 40's degree by 85.53% and cost by 18.54%. The eleven-variable result beats both reviewed package 38 Pareto points; its degree reduction against the strongest degree comparator is 16.32%. Natural-domain nine-variable translations have the same degrees and three more operations. The eleven-positive two-equation system has maximum degree 952 and 108 operations after transport; the twelve-positive three-equation system has maximum degree 888 and 102 operations after transport.

There is no single numerical simplicity order. The nine-variable result is recommended when the number of positive unknowns matters. For unrestricted integers, retain the ten-variable Pareto alternatives rather than hiding the cost–degree tradeoff. The eleven- and twelve-positive equations are useful when low total degree and short evaluation are preferred. The historical 67-operation scalar and 59-operation system remain unbeaten here on their operation measures alone.

1.1 New mechanisms

1. Isolate only the 71 active quadratic monomials from all 528 possible monomials, allowing irrelevant collisions. Overlapping target bands and an interior padding position further shorten the source.
2. In the nine-variable norm, infer a positive margin before recovering integer roots. That margin itself separates the radicands, eliminating a large multiplicative separator.
3. Use a second-order Pell square-root majorant whose polynomial defect has lower degree, and absorb an entire divisibility quotient by an exact CRT reparameterization of an existing witness.
4. Evaluate the sparse source by homogenizing the shared sixty-gate parent and adding only its diagonal correction, instead of expanding every coefficient independently.
5. For eleven positive variables, combine a lean Pell carrier with a smaller absorption modulus Y, proving the missing quotient only after a short Pell-index interval is recovered.

The smaller compiler tuple is reused as a *whole* from packages 39/44; its coefficient size is not a new record. The support certificates are explicit verified constructions, not optimality certificates.

2 Source snapshot and logical dependencies

The construction snapshot is `124c07c0b77a9e20e84676514d8b1ff78536ea3f`. The package-local sources 38–45, not the stale collection summary, provide the reviewed comparisons. At this pin the root catalogue still describes wave 5, and the incoming README says its queue is clear through batch 221. We retained 42 byte-exact files in the current review ledger. Independent gate traversals check the important 38/44 comparison variants; their degree claims are read with their ordinary proofs, rather than presented as newly formalized results.

The proof chain is

$$\begin{aligned}\text{a zero of a displayed fixed polynomial} &\longleftrightarrow \text{a zero of the positive quadratic parent}\\ &\longleftrightarrow \text{blank K215 halting}\\ &\longleftrightarrow \text{a finite syntactic ZFC refutation.}\end{aligned}$$

The sparse source, new algebraic finalizer arguments and degree proofs are proved here. The normalized native compiler, its actual-machine calibration, the Jones ratio bridge, and the relevant Sun ten-integer bridge remain identified mathematical dependencies. Their source proof texts and exact pins are included as inert references. No upstream program is executed by the release verifier.

2.1 The complete quadratic parent

The 31 strictly positive coordinates, in certificate order, are

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_a, \Delta.$$

The fixed ports are β, K_0, M_C, M_F. Define polynomial abbreviations

$$q = \beta J + 1, \quad k = \eta + \zeta, \quad H = 4a + 3, \quad C = q - F - \alpha,$$
$$U = L + 1, \quad Y = a - E, \quad Q = c - \eta, \quad T = P - k, \quad R = jc - V.$$

The following are all 22 equations, in their canonical order:

$$1: \quad U = q^2, \qquad\qquad 2: \quad N = qU,$$

$$
\begin{aligned}
3:&\quad X = wq, & 4:&\quad Y = sN,\\
5:&\quad E = XY, & 6:&\quad Q = kY,\\
7:&\quad T = EQ, & 8:&\quad \tau^2 = PT + 1,\\
9:&\quad D = X + ac + \gamma H, & 10:&\quad \Delta = a^2 + H,\\
11:&\quad C_2 = c^2, & 12:&\quad D^2 = \Delta C_2 + 1,\\
13:&\quad k = R + 1 + hE,\\
14:&\quad R = (q - F)(N - q) - CL + (M_C + qM_F)J,\\
15:&\quad (K_0 + w)C = F + z\beta J,\\
16:&\quad F_2 = f^2, & 17:&\quad V = of - c,\\
18:&\quad g = y^2 - V^2, & 19:&\quad I = iC_2,\\
20:&\quad B_a = I^2, & 21:&\quad B_a = \Delta(F_2 - 1),\\
22:&\quad y^2 - B_a g = 1.
\end{aligned}
$$

The residual is left minus right. Substitute every positive coordinate by $n_i + 1$ to obtain $r_j(\mathbf{n})$ over $\mathbb{N}^{31}$. In particular $r_5(\mathbf{0}) = 1$, so a simultaneous zero cannot be the zero vector. The parent JSON is a literal shared sixty-operation evaluator, with SHA-256

`ac2a67782d344d1e74fd93bc3f09fae0097eaed1829fd8194ae235ae84763e48`.

Its old embedded instance label is historical: the port-parametric algebra is instantiated here by the complete new tuple. Its transport convention $z_{\mathrm{old}} = z + 1$ is valid on solution sets, not a bijection on arbitrary positive assignments. The authentic compiler gives $C > 0$, $F < q$, $X = wq = 2^R$ and $R > 3q$, hence $w > q > F$; therefore $(z_{\mathrm{old}} - 1)(q - 1) = (K_0 + w)C - F > 0$. Thus $z_{\mathrm{old}} \geq 2$, which justifies the inverse transport for the new tuple too.

2.2 Actual fixed integers

The active tuple is `wang943-complement67-shared-anchor-length8807-prime581197`. Its native base is 2^{590069} and $\beta = 2^{590069} - 1$. In port order (β, K_0, M_C, M_F), the bit lengths are

$$(590069, 592884, 590069, 590070).$$

All four minimal unsigned little-endian payloads are shipped, both raw and as deterministic gzip, totaling 295388 raw or 9445 compressed bytes. M_F is the already-shifted `MF_source`; it must not be shifted again. The manifest gives every SHA-256 and the exact alias map. Derived leaves are materialized separately for the nine- and ten-variable circuits. These are ordinary fixed integer coefficients, not existential parameters, unknown enumerator indices, or instructions to wait for a consistency decision.

An independently authored checker reconstructs the four ports from the literal 943-tile data and the pinned complement-decoder formulas. It verifies 66062 scalar basis identities, 38663 single-bit mutations, synchronization supports, carry inequalities, mask balance, controller primality, exact orders, and lifting/unit conditions. The source expansion contains 136 weighted coefficient records. Writing $c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i! \, [\mathbf{n}^\alpha] r_j$, the checked bounds are

$$|c_{j,\alpha}| < M = 2^{1182955}, \qquad \text{bitlength}\left(1 + \sum |c_{j,\alpha}|\right) = 1182954.$$

The nine-variable radix scale is $\lambda = 4096M = 2^{1182967}$; the integer source uses $\beta_S = 2^{1182966} > 2048(1 + \sum |c_{j,\alpha}|)$. These finite coefficient checks do not replace the universal compiler proof.

3 Sparse active isolation and overlapping target bands

The full B_2 requirement is unnecessarily strong. The shifted quadratic source has only 71 distinct active monomials, although its universal quadratic expansion has 528. We require each active monomial to be isolated from that expansion, but allow all irrelevant monomials to collide.

Let $q_j = \sum_\alpha p_{j,\alpha} X^\alpha$ be the shifted residuals, $w(\alpha) = \sum_i \alpha_i w_i$, and $f(\alpha) = (2 - |\alpha|)! \prod_i \alpha_i!$. Put

$$C_{\mathbf{z}}(T) = \left(1 + \sum_i z_i T^{w_i}\right)^2, \qquad D(T) = \sum_{j,\alpha} f(\alpha) p_{j,\alpha} T^{H_j - w(\alpha)}.$$

All source exponents are nonnegative. Our exact finite separation condition is

$$H_j - w(\alpha) + w(\gamma) = H_k \quad \Longleftrightarrow \quad j = k \text{ and } \alpha = \gamma, \tag{S}$$

for every active source record (j, α), every monomial γ of degree at most two, and every target H_k.

Theorem 3.1 (Sparse simultaneous extraction)**.** *Under* (S)*, for every integer tuple* $\mathbf{z}$*,* $[T^{H_j}](C_{\mathbf{z}} D) = 2q_j(\mathbf{z})$ *for every* j*. Moreover, distinct source records occupy distinct exponents in* D*. If every weighted source coefficient is bounded in absolute value by* M*, then every coefficient of* D *is so bounded, and for nonnegative* $\mathbf{z}$ *every coefficient of* $C_{\mathbf{z}} D$ *is bounded by* $M(1 + \sum_i z_i)^2$*.*

Proof. Expand the square retaining its formal multinomial labels, even if some irrelevant labels have the same exponent. Condition (S) permits only the intended label at a target, whose multinomial multiplier $2/f(\alpha)$ cancels the weighting in D. If two source records occupied the same exponent, multiplying by either record's monomial would violate (S) at its target. Hence source exponents are distinct. Finally the coefficients of $C_{\mathbf{z}}$ are nonnegative and sum to $(1 + \sum_i z_i)^2$, giving the asserted convolution bound. □

The certified variable weights, in source order, are

$$(9, 89, 38, 43, 2, 118, 82, 27, 88, 83, 23, 163, 123, 148, 3, 108,\\ 144, 155, 137, 173, 174, 68, 17, 107, 63, 57, 169, 168, 28, 133, 102).$$

Their maximum is 174, versus 799 previously. The 528 monomials occupy only 265 positions: this is deliberately not a Sidon set. All 71 active monomials are nevertheless isolated. With rows ordered $1, 2, 3, 4, 6, \ldots, 22, 5$, regular targets $H_j = 349j$ yield $D_{\max} = 7678$ and $K_{\text{code}} = 8026$.

An independently checked overlapping placement has targets

$$(63, 3297, 4744, 1100, 752, 3094, 2209, 3586, 912, 4689, 2437,\\ 4168, 3833, 2764, 78, 4364, 425, 1347, 1548, 1614, 1882, 5074),$$

yielding $D_{\max} = 5074$ and $K_{\text{code}} = 5422$. In both cases the largest target belongs to shifted row 5, whose constant coefficient is 1, so D has leading coefficient 2 and is positive at all sufficiently large positive bases. The existing target mask uses only distinctness of its target positions, not their equal spacing. Thus the same no-carry theorem applies without an extra factor 22 in its coefficient bound. Both layouts are retained because irregular exponent arithmetic can trade lower degree for more operations.

The supplied standard-library verifier reconstructs the source symbolically and checks all 71 808 source-term/expansion pairs. The search itself is not trusted and no global optimality is claimed. An additional certified layout puts the padding position at 81, with one variable position at 0, and replaces $(1 + g)^2$ by $(B^{81} + g)^2$. Its packed layout has $D_{\max} = 4728$, $K_{\text{code}} = 5076$, and largest target 4890. The leading coefficient remains 2. The decoding and coefficient bounds are unchanged. Completeness still gives $g \geq b$:

no source zero can be supported on its sole zero-position variable i, as shifted row 5 is independent of i and has constant 1. All bounds use $D_{\max}$ rather than the largest target, which is different for interior padding. Full vectors and a checked translation-normalization certificate are included alongside the general proof. For the wider nine-variable witness interval $0 \le g < 4bB^W$, $B^{81}+g < (4b+1)B^W \le 5bB^W$; for Sun's interval $0 \le g < 2bB^W$, the corresponding bound is $3bB^W$. Thus interior padding preserves the respective coefficient budgets. Uniformly raising all packed interior targets by 186 makes their minimum $348 = 2W$, permitting a shared homogeneous evaluator with no negative powers. This multiplies D by T^{186}, preserves every extraction identity, and gives $D_{\max} = 4914$ and $K_{\text{code}} = 5262$. It is supplied as a separately verified degree–operation tradeoff.

3.1 Complete small vectors for the retained certificates

All layouts use row order $(1, 2, 3, 4, 6, 7, \dots, 22, 5)$. The interior padding is $p = 81$ and its variable weights, in the 31-coordinate order already displayed, are

$$(51, 169, 164, 28, 174, 146, 0, 32, 20, 24, 154, 45, 143, 140, 33, 141,$$
$$44, 123, 135, 11, 119, 148, 145, 18, 166, 1, 158, 25, 163, 29, 5).$$

For the packed interior variant the target vector is

$$(4129, 3084, 1260, 1688, 1956, 1467, 2840, 2579, 3568, 3318, 4352,$$
$$4608, 483, 793, 3980, 1039, 2332, 3719, 3313, 2101, 162, 4890).$$

The shifted packed variant adds 186 to every target and every encoding exponent. The regular interior variant uses $h_j = 349j$ for $1 \le j \le 22$. These operations specify their complete target vectors without ellipses that hide any irregular data. The following parameters distinguish the largest target from the largest encoding exponent:

Layout	p	$\max h_j$	d_c	K_c
Original regular, $W = 174$	0	7678	7678	8026
Original packed, $W = 174$	0	5074	5074	5422
Interior regular, $W = 174$	81	7678	7516	7864
Interior packed, $W = 174$	81	4890	4728	5076
Interior shifted packed, $W = 174$	81	5076	4914	5262
Reference regular, $W = 200$	0	8822	8822	9222
Reference packed, $W = 200$	0	5808	5808	6208

The two $W = 200$ certificates are retained as checked search comparison data, not recommendations. Their weights and packed targets are listed below; the regular targets are $h_j = 401j$. Weights:

$$(14, 37, 200, 110, 183, 141, 125, 174, 61, 45, 86, 158, 189, 84, 198, 165,$$
$$46, 115, 173, 36, 19, 44, 108, 12, 126, 3, 167, 150, 182, 135, 4).$$

Packed targets:

$$(5194, 3836, 1868, 2654, 3309, 1246, 3014, 4769, 1802, 72, 4141,$$
$$650, 2229, 941, 2605, 4531, 5570, 3585, 5138, 379, 1547, 5808).$$

Every certificate also contains all 136 exact coefficient recipes and source exponents. The standard-library verifier reconstructs the parent independently, rather than trusting these lists.

4 The nine-variable conditional-separation construction

We first work in the mixed domain

$$b, g, h, s, w, i \in \mathbb{Z}_{>0}, \qquad \phi, j, n \in \mathbb{N}.$$

These are exactly nine unknowns. Every other name below is a computed polynomial, not an additional quantified variable. Use any supplied valid packing certificate, in particular the interior packed certificate for the lowest degree. Let p be its padding position, W its maximum position, $D_c(T)$ its weighted code, $d_c = \deg D_c$ and $K_c = d_c + 2W$.

4.1 A quadratic radix with unconditional positivity

Set $\lambda = 2^{1182967}$ and define

$$B = \lambda b^2, \qquad \ell = \sum_i B^{w_i}, \qquad \mathcal{T} = \sum_j B^{h_j},$$

$$N_0 = B^{W+1}, \qquad N_1 = 4B^{K_c+1}, \qquad N = N_0 N_1,$$

$$M_c = N_0 - 1 - (b-1)\ell,$$

$$T_c = M_c + (B-2)\mathcal{T} N_0,$$

$$J_c = 2(B^p + g)^2 D_c(B) + B \sum_{e=0}^{K_c} B^e,$$

$$S_c = g + J_c N_0,$$

$$R = (N-1)\big[(S_c + T_c + 1)N + T_c + 1\big],$$

$$G_c = 32 b B^W.$$

Every exponent and sum limit is a fixed nonnegative integer. The scale is an ordinary fixed coefficient; there is no enormous additive floor on b. Since D_c has leading coefficient 2 and all lower coefficients have absolute value below $M = 2^{1182955}$, while $B - 1 > M$,

$$D_c(B) > B^{d_c} > 0, \qquad D_c(B) < \frac{MB^{d_c+1}}{B-1}.$$

Also $M_c > 0$. Thus the registers satisfy $R \geq N \geq 8$ and $0 < b \leq N \leq R$, unconditionally, before any polynomial zero is assumed.

On the branch $g < 4bB^W$, interior padding needs the estimate $B^p + g < (4b+1)B^W \leq 5bB^W$, rather than the old zero-padding estimate. Consequently

$$2(B^p + g)^2 D_c(B) < \frac{50Mb^2 B^{K_c+1}}{B-1} \leq \frac{B^{K_c+2}}{64(B-1)}.$$

The geometric summand is less than $2B^{K_c+1}$, proving $0 < J_c < N_1$. Moreover $4b < B$ gives $g < N_0$, and the distinct targets satisfy $\max h_j < K_c$, so $0 \leq T_c < N$ and $0 \leq S_c < N$.

Lemma 4.1 (Exact binary carry identity)**.** *For $N = 2^r$, $0 \leq S, T < N$, and $X = (N-1)((S+T+1)N + T + 1)$,*

$$N^2 \mid \binom{2X}{X} \quad \Longleftrightarrow \quad \textit{the binary addition of } S \textit{ and } T \textit{ has no carries.}$$

Proof. The exact identity $X = (S+T)N^2 + (N-1-S)N + (N-1-T)$ has disjoint lower blocks even if $S+T \geq N$. Therefore

$$\mathrm{pc}(X) = \mathrm{pc}(S+T) + 2r - \mathrm{pc}(S) - \mathrm{pc}(T) = 2r - \tau_2(S,T),$$

where τ_2 is the total number of binary carries. Legendre's formula says $v_2\binom{2X}{X} = \mathrm{pc}(X)$, proving both directions. □

When b is a power of two, so are B, N_0, N_1, N. Lemma 4.1 splits the binomial divisibility into the low and high block carry tests. The low mask has digit $B-b$ at every variable position, and $B-1$ at every other position. It therefore forces exactly

$$g = \sum_i z_i B^{w_i}, \qquad 0 \leq z_i < b.$$

It excludes the distinct padding position, so the coefficient of T^p in the encoded linear form remains 1. For this decoded vector, every coefficient of $(T^p + \sum_i z_i T^{w_i})^2 D_c(T)$ has absolute value below $M(1+\sum_i z_i)^2 \leq M(32b)^2 = B/4$. Thus

$$J_c = 2\sum_{e=0}^{K_c} a_e B^e, \qquad a_e = [T^e]\big((T^p + \sum_i z_i T^{w_i})^2 D_c(T)\big) + B/2,$$

with $0 < a_e < B$. Dividing both high-block operands by 2 removes a common binary shift. At each target, the mask becomes $B/2-1$, whose lower bits are all 1; its addition to a_{h_j} is carry-free exactly when $a_{h_j} = B/2$. Sparse extraction makes this equivalent to $r_j(\mathbf{z}) = 0$. The other positions impose no condition.

Conversely, from a source zero choose a power of two b greater than every coordinate, as large as needed, and use the displayed digit encoding for g. Distinct positions give $g < bB^W$. The source zero is nonzero, so $g > 0$; all carries vanish and $32g < G_c$. This stronger bound supplies completeness of the inherited numerical ratio theorem.

4.2 Literal ratio registers and exact quotient removal

Define

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M_0 &= RY,\\
P &= 2M_0^2 U, & K &= R+1+h(P-1), & A &= M_0(U+1),\\
B_1 &= 2R+1, & W_1 &= bw, & C &= B_1 + W_1 + \phi,\\
J &= 4A-5, & Q &= 3W_1 C - 2(W_1^2-1), & &\\
D &= (A^2-1)C^2+1, & k &= iJC, & L &= kD,\\
E &= LC, & F &= (A^2-1)E^2+1, & &\\
G &= A + F(CD+1-A), & H &= C + F(B_1+jC), & &\\
I &= (G^2-1)H^2+1, & a_0 &= (P^2-1)K^2+1, & a_1 &= DFI,\\
m &= [K^2 - 4(C-KY)^2]G_c - 8gK^2. & & & &
\end{aligned}$$

The inherited ratio theorem instead has $H_{\mathrm{old}} = B_1 + j_{\mathrm{old}} C$, $j_{\mathrm{old}} \geq 1$, and divisibility $JF \mid QF + J(H_{\mathrm{old}} - C)$. Since $F \equiv 1 \pmod J$ and $F \equiv 1 \pmod C$, $\gcd(F,J) = 1$, so that divisibility is equivalent to

$$J \mid Q \quad \text{and} \quad F \mid H_{\mathrm{old}} - C.$$

If the second holds, $z = (H_{\text{old}} - C)/F$ is positive and congruent to B_1 modulo C. As $0 < B_1 < C$, uniquely $z = B_1 + jC$ for $j \geq 0$. This is exactly the new H. Conversely every such new j gives the old positive integer

$$j_{\text{old}} = 1 + \frac{F-1}{C} B_1 + Fj.$$

The quotient is an integer because $C \mid F - 1$. This is a bijection of the old divisible witnesses with the new unrestricted nonnegative witnesses, preserving H itself. No cofinal-witness assertion or extra hypothesis is introduced. The resulting ratio conditions are precisely

$$a_0 \text{ square}, \quad a_1 \text{ square}, \quad J \mid Q, \quad m > 0.$$

The numerical Jones theorem gives from these conditions that b is a power of two, $8g < G_c$, and $N^2 \mid \binom{2R}{R}$. Conversely the stronger source bound $32g < G_c$ produces the requisite ratio witnesses.

4.3 A general conditional-separation norm theorem

Theorem 4.2. *Let $u, v, m, a_0, a_1, V_0, V_1$ be integers with $u > 0$, $0 < v/u < V_0 + V_1$, $a_i > 0$, $V_i > 0$, $a_i < V_i^2$, and $m > 0 \Rightarrow a_0 \neq a_1$. Put*

$$\Omega = um, \qquad \Theta = un + v - \Omega(V_0 + V_1).$$

Then there exists $n \in \mathbb{N}$ satisfying

$$(\Theta^2 - \Omega^2(a_0 + a_1))^2 - 4\Omega^4 a_0 a_1 = 0$$

if and only if $m > 0$, a_0, a_1 are integer squares, and $u \mid v$.

Proof. If $m = 0$, $\Theta = un + v > 0$, excluding a zero. Otherwise factor the norm over the reals to obtain signed roots $r_i^2 = a_i$ with $\Theta/\Omega = r_0 + r_1$. Rearrangement gives

$$n + v/u = mZ, \qquad Z = V_0 + V_1 + r_0 + r_1 > 0.$$

The left side is positive, so $m > 0$ *before* integrality of either root is known. Conditional separation now makes $x = r_0 + r_1$ nonzero; otherwise $a_0 = a_1$. Since x is rational, $2xr_0 = x^2 + a_0 - a_1$ shows $r_0, r_1 \in \mathbb{Q}$. A rational number whose square is an integer is an integer, as follows by writing it as a reduced fraction. Hence $Z \in \mathbb{Z}$ and $u \mid v$. Conversely choose the positive integer roots and put $n = mZ - v/u$. Then $n \in \mathbb{Z}$ and $n \geq Z - v/u > V_0 + V_1 - v/u > 0$. Substitution makes a linear factor zero. The quotient bound is needed only for completeness, not for the positivity-first soundness argument. □

Positive majorants are essential, not just $V_i^2 > a_i$. Without separation, $a_0 = a_1 = 2$, $V_0 = V_1 = 2$, $u = v = 2$, $m = 1$, $n = 3$ is a false square witness. Without positive majorants, $a_0 = 1, a_1 = 4, V_0 = -2, V_1 = 4$, $u = v = 1, m = -1, n = 0$ is a false positive-margin witness, even though the squared-majorant and quotient inequalities hold. These are retained as negative controls.

4.4 Sharper Pell majorant and the actual equation

Unconditionally $A \geq 2$ and $P, K, C, J, D, E, F, G, H > 1$. Define

$$V_0 = PK, \qquad T = k(8D^2 - 4D - 1), \qquad V_1 = TGH.$$

The identities $E = CL$, $L = kD$ give $F = (D-1)k^2D^2 + 1$. Direct expansion then yields

$$V_0^2 - a_0 = K^2 - 1 > 0, \qquad T^2 - 64DF = (8D+1)k^2 - 64D = 8D(k^2 - 8) + k^2 > 0,$$

because $k = iJC \geq 3$. Therefore the strictly positive majorant V_1 has

$$\Delta_1 = V_1^2 - 64a_1 = [(8D+1)k^2 - 64D](GH)^2 + 64DF(H^2-1) > 0.$$

The lower degree of this polynomial defect, rather than simply a smaller numerical bound, is the degree improvement.

If $m > 0$, its defining margin implies $4(C - KY)^2 < K^2$, so $C > K(Y - 1/2) > K$ because $Y \geq 64$. Also

$$A^2 - P = M_0^2(U^2+1) > 0, \quad D > C^2, \quad F \geq A^2, \quad G \geq A, \quad I \geq G^2 \geq A^2.$$

For $G \geq A$, use $G = A + F(CD + 1 - A)$ and $D \geq A^2$, $C \geq 1$. It follows that

$$a_1 = DFI > C^2A^4 > C^2P^2 > K^2P^2 > a_0.$$

This separation is established without assuming squarehood or divisibility. The same positivity-first proof works without the auxiliary quotient offsets. Indeed $Q < 3C^2$, $D > C^2$ and $T > 3D$ imply $V_1 > Q$. The promised fixed polynomial is

$$\begin{aligned} \Omega &= Jm, \\ \Theta &= Jn + Q - \Omega(V_0 + V_1), \\ F_9 &= [\Theta^2 - \Omega^2(a_0 + 64a_1)]^2 - 256\Omega^4 a_0 a_1 = 0. \end{aligned}$$

To check the offset-free form explicitly, $m = 0$ gives $\Theta = Jn + Q > 0$. Otherwise a zero factor gives $n + Q/J = mZ$ with $Z = V_0 + V_1 + r_0 + r_1 > 0$. Thus $m > 0$, and the separation just proved makes $r_0 + r_1$ nonzero. The rational reconstruction from Theorem 4.2 gives integer roots and $J \mid Q$. Conversely choose the positive integer roots and set $n = mZ - Q/J$. As $Z > V_1 > Q > Q/J$, this is a positive integer and a factor vanishes.

Squarehood of $64a_1$ is equivalent to squarehood of a_1: an integer square root divided by 8 is a rational square root of an integer, hence an integer. The finalizer is therefore equivalent to exactly the four ratio conditions. Composing both directions proves the desired source solvability equivalence.

For the positive-domain polynomial, replace ϕ, j, n in this mixed-domain formula by the corresponding positive coordinates minus 1. For the nonnegative-domain polynomial, replace b, g, h, s, w, i by the corresponding nonnegative coordinates plus 1. These are bijections; the supplied circuit ledgers charge their operations. They must not be omitted when stating the domain.

4.5 Exact degree and the retained evaluator frontier

Let $\ell_0 = K_c + W + 2$ and $t = 2\ell_0 = \deg N$. On the all-variable diagonal, the leading coefficients of N, R are respectively $4\lambda^{\ell_0}$ and $16\lambda^{3\ell_0}$. The geometric summand uniquely dominates J_c. Direct propagation gives

Register	Degree	Register	Degree
P	$12t + 3$	K	$12t + 4$
A, J	$7t + 2$	C	$3t$
D	$20t + 4$	k	$10t + 3$
L	$30t + 7$	E	$33t + 7$
F	$80t + 18$	G	$103t + 22$
H	$83t + 19$	I	$372t + 82$
a_0	$48t + 14$	a_1	$472t + 104$
m	$28t + 2W + 11$	Ω	$35t + 2W + 13$
V_0	$24t + 7$	V_1	$236t + 52$
Δ_1	$412t + 92$		

The Δ_1 degree follows from its positive decomposition, whose first summand uniquely dominates. Put $E_0 = Jn + Q$. The exact identity

$$\begin{aligned}\Theta^2 - \Omega^2(a_0 + 64a_1) = E_0^2 - 2E_0\Omega(V_0 + V_1) \\ + \Omega^2[(K^2 - 1) + \Delta_1 + 2V_0V_1]\end{aligned}$$

resolves the leading cancellation. The uniquely dominant term is $\Omega^2\Delta_1$, of degree $482t + 4W + 118$. Its square dominates the subtracted norm term. Thus

$$\deg F_9 = 964t + 8W + 236, \qquad \mathrm{lc}_{\mathrm{diag}}(F_9) = 2^{1730}\lambda^{964\ell_0+4W}.$$

This is nonzero for the actual tuple, and a multivariate upper bound matches this diagonal lower bound, proving exact total degree.

The regular interior layout costs 548 operations at degree 15502748. Uniformly shifting the packed interior targets by 186 makes their minimum $2W$, allowing the shared homogeneous source evaluator described in the next section: 603 operations at degree 10486092. The unshifted packed layout has the lowest degree, 10127484, at 655 operations. The natural-domain forms cost three additional operations each. Larger equal-polynomial degree-certificate circuits are not substituted for these primary operation counts.

5 The ten-unrestricted-integer companion

The sparse quadratic code can be composed with the existing ten-integer finalizer, without a new unknown or a change to its logical interface. The variables are exactly

$$(f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}^{10}.$$

The coefficient tuple is the coherent tuple specified in the coefficient manifest. All expressions below are polynomial abbreviations, not extra quantified variables. Our new work is sparse support isolation, an interior padding digit, and a shared-source evaluator. The universal ten-integer reduction itself is due to Sun.

5.1 A source interface with arbitrary padding position

Let p be the padding position and $w_1, \ldots, w_{31}$ the distinct variable positions. They are distinct from p, nonnegative, and at most W. For $|\alpha| \le 2$ put

$$e_p(\alpha) = (2 - |\alpha|)p + \sum_i \alpha_i w_i, \qquad c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i!\, p_{j,\alpha},$$

where $r_j(n) = \sum p_{j,\alpha} n^\alpha$ is the shifted positive parent. The certificate gives targets H_j such that

$$H_j - e_p(\alpha) + e_p(\gamma) = H_k \quad \Longleftrightarrow \quad (j, \alpha) = (k, \gamma)$$

for each nonzero source coefficient and each quadratic monomial γ. In particular,

$$[T^{H_j}] \left(T^p + \sum_i n_i T^{w_i} \right)^2 \sum_{k,\alpha} c_{k,\alpha} T^{H_k - e_p(\alpha)} = 2r_j(n).$$

This condition is weaker than requiring all quadratic monomial positions to be distinct: only the source's active support must be isolated. Write

$$D(T)=\sum_{j,\alpha}c_{j,\alpha}T^{H_j-e_p(\alpha)},\quad Z(T)=\sum_j T^{H_j},\quad \ell(T)=\sum_i T^{w_i},\quad K_c=\deg D+2W.$$

All the exponents displayed are certified nonnegative.

The exact fixed integer $C_0=1+\sum_{j,\alpha}|c_{j,\alpha}|$ has bit length 1 182 954. Independent reconstruction verifies

$$\beta_S=2^{1\,182\,966}>2048C_0.$$

In particular the fixed base multiplier is 1 modulo 3. Define

$$\begin{aligned}
b&=1+3f, & \mathcal{B}&=\beta_S b^2, & \mathcal{C}&=b\mathcal{B}^W,\\
N_0&=\mathcal{B}^{W+1}, & N_1&=4\mathcal{B}^{K_c+1}, & N&=N_0N_1,\\
M_c&=N_0-1-(b-1)\ell(\mathcal{B}),\\
T_c&=M_c+N_0(\mathcal{B}-2)Z(\mathcal{B}),\\
J_c&=2(\mathcal{B}^p+g)^2D(\mathcal{B})+\mathcal{B}\sum_{i=0}^{K_c}\mathcal{B}^i,\\
S_c&=g+N_0J_c, & R_c&=(S_c+T_c+1)N+T_c+1,\\
X&=(N-1)R_c, & Y&=N^2.
\end{aligned}$$

Suppose $f\neq 0$, b is a square and $0\le g<2\mathcal{C}$. Then $b\ge 4$ and $\mathcal{B}>2048C_0b^2$. Since $p\le W$,

$$\left|2(\mathcal{B}^p+g)^2D(\mathcal{B})\right|<18C_0b^2\mathcal{B}^{K_c}<\mathcal{B}^{K_c+1}.$$

Hence $0<J_c<3\mathcal{B}^{K_c+1}<N_1$ and the block bounds give $0\le S_c,T_c<N$. Consequently

$$X\ge N^2-1>3b,\qquad Y>\max\{b,256\},\qquad 3\mid X.$$

These statements precede, and do not assume, the power-of-two condition.

If b is also a power of two, Sun's binomial carry lemma identifies $Y\mid\binom{2X}{X}$ with the absence of binary carries in S_c+T_c. The low block decodes $g=\sum_i n_i\mathcal{B}^{w_i}$ with $0\le n_i<b$. Every coefficient A_t of $(T^p+\sum n_iT^{w_i})^2D(T)$ satisfies

$$|A_t|\le C_0(1+\textstyle\sum n_i)^2<1024C_0b^2<\mathcal{B}/2.$$

Thus the baseline $\mathcal{B}/2$ produces actual, strictly interior base-$\mathcal{B}$ digits, without interdigit carrying. At target H_j, the digit is $\mathcal{B}/2+2r_j(n)$. Its addition to $\mathcal{B}/2-1$ has no binary carries precisely when $r_j(n)=0$. This proves soundness.

Conversely, choose a natural source zero, then choose $b=4^q$ arbitrarily large and greater than all its coordinates. The same calculation gives the binomial divisibility and $g<\mathcal{C}$. In the interior-padding certificate the sole zero variable position is the coordinate i, whereas the fifth residual has constant term 1 and depends only on E,a,X, whose positions are positive. Some positive-position coordinate must therefore be nonzero. Thus $g\ge\mathcal{B}\ge b$. The original padding-zero certificate has all variable positions positive, and the same conclusion follows because the zero vector fails the fifth residual. This proves the full source interface required by the inherited bridge.

5.2 The literal tail

The following formulas explicitly define the tail. Their number-theoretic justification is the cited Sun bridge and the previously proved coprime positivity merger, quintic divisibility merger and separating norm.

$$L = lY, U = 2LX, V = 4gwY, W_1 = bw,$$
$$A = U(V+1), B_1 = 2X+1, C = B_1 + (A-2)h,$$
$$P_n = U^2V, K_1 = X + 1 + k(P_n - 2),$$
$$S = 2A - 5, T = 3W_1C - 2(W_1^2 - 1),$$
$$r_1 = (P_n^2 - 4)K_1^2 + 4,$$
$$\Delta = 8\mathcal{C}^3 g K_1^2 - g^2(32(C - K_1L)^2\mathcal{C}^3 + g^2K_1^2),$$
$$O = f^2l^2\Delta, \xi = zx,$$
$$D = (A^2-4)C^2 + 4, E = C^2D\xi,$$
$$F = 4(A^2-4)E^2 + 1, G = 2(A+2)(A-2)^2E^2 - 1,$$
$$H = C + (B_1 - C)F + 16EFy, I = (G^2-1)H^2 + 1,$$
$$r_0 = b(1 + (3O-4)b^2z^2), r_2 = DFI,$$
$$\Sigma = \xi S^5, t = T + Sy,$$
$$\Theta = (t^3 - tS^2 - S^3)(t^2 + 23S^2),$$
$$q_0 = 1, q_1 = r_0, q_2 = r_0(P_n^2 + K_1^2 + 3),$$
$$a_j = \Sigma^2 q_j^2 r_j \quad (0 \le j \le 2), Z = \Sigma m - \Theta.$$

Finally set

$$V_n = Z^2 + a_2 - a_0 - a_1, \qquad W_n = V_n^2 + 4Z^2a_2 - 4a_0a_1,$$
$$\boxed{F_{10} = W_n^2 - 16V_n^2Z^2a_2 = 0.} \tag{1}$$

There is no radical, division, variable exponent, unspecified enumeration index or hidden existential witness in this equation.

The final polynomial is the formal norm $\prod_{\varepsilon\in\{\pm1\}^3}(Z + \sum_j \varepsilon_j\sqrt{a_j})$. The inherited separating-weight proof applies even to candidate negative radicands, and the $\Sigma = 0$ branch is excluded because then $F_{10} = \Theta^8 \neq 0$. No claim of an unweighted three-radical criterion is being made. Composing that proof with the source interface just proved and the coherent compiler endpoint yields

$$\mathrm{ZFC} \vdash (\neg\exists \mathbf{z} \in \mathbb{Z}^{10} : F_{10}(\mathbf{z}) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

This is an equivalence theorem, not a proof of consistency or inconsistency.

5.3 Two layers of evaluator factoring

The tail polynomial is unchanged by the exact identities

$$G = (a-1)(F-1) - 1, \quad a = (LX)(V+1), \qquad H = C + F(B_1 - C + 16Ey),$$
$$t^3 - tS^2 - S^3 = t(t^2 - S^2) - S^3, \qquad f^2l^2 = (fl)^2, \quad b^2z^2 = (bz)^2.$$

The equality $A = 2a$ is built polynomially, without division. Sharing $\Sigma^2 r_0^2$ between a_1 and a_2 saves another operation. Independent symbolic expansions verify these identities and the norm.

There is a more substantial source-level saving. Let $Q_j(u_0, \dots, u_{31})$ be the degree-two homogenization of the shifted residual, and let $\mathrm{diag}(Q_j)$ be its sum of pure-square monomials. The weighted form is exactly

$$V_j = Q_j + \mathrm{diag}(Q_j).$$

Indeed a degree-two multinomial has weight 1 on cross monomials and weight 2 on pure squares. Evaluate every Q_j by homogenizing the *shared sixty-gate parent*, not by expanding its monomials. At

$$u_0 = \mathcal{B}^{W-p}, \qquad u_i = \mathcal{B}^{W-w_i},$$

we have $D(\mathcal{B}) = \sum_j \mathcal{B}^{H_j-2W} V_j$ whenever every $H_j \geq 2W$. Only the sparse diagonal correction is added afterward. This preserves sharing between different quadratic equations and sharply reduces the operation count. Uniformly translating all target positions preserves their isolation property, so it can be used to ensure this nonnegative-offset condition.

All operation counts are live binary $+, -, \times$ gates, including scalar multiplication; fixed coefficients and reuse are free. The supplied JSON circuits and receipts give the exact counts for each retained tradeoff. Exact degree is established by an upper bound from circuit propagation and a nonzero coefficient at that same bound after diagonal specialization modulo a prime. Three independent primes are retained for redundancy. The verifier also checks actual fixed bytes, all coefficient bounds and literal source-and-tail replay. These finite checks do not replace the inherited unbounded arithmetic theorems, nor supply a new end-to-end formal proof assistant development.

6 A new lean Pell carrier: eleven positive variables

Result. There is an explicit polynomial in eleven strictly positive integer variables of exact total degree 954, evaluated by 113 binary arithmetic operations, whose absence of positive solutions is equivalent in ZFC to Con(ZFC). The associated two-equation system has maximum degree 952 and costs 108 operations after a valid positive transport-coordinate shift. A twelve-positive companion has degree 897 and 108 operations, or a three-equation system of maximum degree 888 and 102 operations. These are ordinary mathematical reductions using the inherited normalized compiler and calibrated K215 machine, not new end-to-end proof-assistant certificates or proofs of consistency.

Compared with the pinned eleven-positive degree-1140/129-operation companion, the new equation removes 186 degrees (16.32%) and 16 operations. The new twelve-positive companion strictly improves the pinned degree-901/112-operation equation on both measures.

Fixed coefficients and explicit equation. Use the entire coherent four-port tuple (β, K_0, M_C, M_F) bound in the supplied coefficient manifest, with $\beta = 2^{590069} - 1$ and M_F already shifted by β. All four ports are fixed integer numerals. They are not existential parameters and may not be mixed with ports from another tuple. The eleven positive unknowns are

$$J, F, \alpha, z, i, T, s, w, \eta, \zeta, t.$$

Define computed polynomial abbreviations

$$\begin{aligned}
q &= \beta J + 1, & s_0 &= s + 3q, & C &= q - F - \alpha,\\
X &= wq, & Y &= s_0 q^3, & E &= XY,\\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y,\\
H &= 4a + 3, & \Delta &= a^2 + H, & \ell &= EkY,
\end{aligned}$$

$$\begin{aligned}
A &= 1+\ell(\ell+k), & G &= 1+\Delta c^2, & b_0 &= X+ac,\\
R &= ((q-1)(q-F)+\alpha)(q^2-1)+(M_C+qM_F)J,\\
N_t &= (K_0+w)C+q-F-z(q-1),\\
I &= ic^2, & D &= 1+\Delta I^2, & Q &= (\Delta I)^2,\\
d &= Q-1, & v &= cT, & b &= c+RD,\\
e &= Q(Dv^2+b^2)-1, & B_a &= 2Qvb.
\end{aligned}$$

The new lean carrier and smaller absorption modulus are

$$K = IT, \quad u = k-R-1, \quad S = HY, \quad L = Y(1+HK), \quad W = Yb_0 + H\big[Y(t+b+\ell)+u\big].$$

Set

$$N = \prod_{\substack{g=\pm\sqrt{G}\\ \tau=\pm\sqrt{A}\\ f=\pm\sqrt{D}}} \left[d(W-Lg-S\tau)^2 - S^2e + S^2B_af\right].$$

This is a formal conjugate product in an integer polynomial ring, not an assertion of independent specialized Galois automorphisms. The displayed radicals add no unknowns. The recommended equation is

$$\boxed{N_t(1+N)-1=0.}$$

Every power and conjugate product is replaced by additions, subtractions, and multiplications in the supplied division-free JSON circuit. In particular, the claimed 113 operations include evaluation of every abbreviation.

Why the omitted index quotient is recoverable. The norm obeys a square congruence modulo 4. Thus the product equation forces $N_t = 1$ and $N = 0$. The authentic outer interface supplies

$$q \ge 16, \quad 3q < R < q^4-2, \quad X \ge q, \quad Y > 3R, \quad E > R+1, \quad k \ge 2, \quad kY < c < k(Y+1).$$

No premature assumption $k > E$ is used. For the positive real roots $g = \sqrt{G}$, $f = \sqrt{D}$, $\tau = \sqrt{A}$, the exact identity

$$(Ig)^2 - (cf)^2 = I^2 - c^2$$

and elementary bounds give

$$\frac{b+cf}{d} < \frac{12}{E} < \frac{ic}{8E} < Ig - cf.$$

Every coherent auxiliary root obeys $|y| < b + ITg$, including complex roots. Hence every inner norm root has $|x| < \tau + b + ITg$. A negative main root, on the other hand, would give

$$x = b + ITg + (t+u/Y) + \ell + \frac{b_0+g}{H} > b + ITg + \tau,$$

because $t + u/Y > 0$, $(b_0+g)/H > c/3 > k$ and $\tau < \ell + k$. Actual quadratic conjugation now forces g integral and x rational.

The inherited modulus-Q exceptional-root exclusion uses $1 < A < \Delta I = \sqrt{Q}$, not a large inner separator. It proves that the rational reconstruction denominators are nonzero and recovers integral τ, f. Rational Pell saturation then makes y and x integral. Reducing the carrier identity modulo Y gives

$$Y \mid H(k-R-1).$$

The unchanged signed auxiliary argument recovers main Pell index $p = R$, and the first Pell norm gives $k = 2\psi_{2EY+1}(n)$ with $1 \le n < R$. Now

$$\mu = \frac{Y}{\gcd(Y, H)} = \frac{Y}{\gcd(Y, 3)} > R, \qquad k \equiv 2n \pmod{Y}.$$

It follows that $2n - R - 1 \equiv 0 \pmod{\mu}$; its absolute value is below R, so $2n = R + 1$. As the same recurrence also gives $k \equiv 2n \pmod{E}$, the full omitted quotient

$$h = \frac{k - R - 1}{E}$$

is a positive integer. Dividing the carrier identity by Y then recovers $g \equiv b_0 \pmod{H}$ and a positive $\gamma = (g - b_0)/H$.

The exact bridge to the ordinary signed compiler sets $i_{\text{ordinary}} = \Delta i$. Then $(i_{\text{ordinary}} c^2)^2 = \Delta(f^2 - 1) = Q$, so its strong and auxiliary equations hold literally. The pinned signed theorem permits the resulting signed auxiliary quotient and extracts a blank K215 halt.

Constructive positivity. A canonical compiler witness has $X = 2^R$, $Y > 2^R$ and $R > 3q$, so $s = s_{\text{old}} - 3q > 0$. Set

$$t = \gamma + ITg + \tau + y - b - \ell - hX.$$

The auxiliary Pell equation yields $y \ge b - vf$, while $\tau - \ell > k/3$ and $hX < k/Y < k/3$. Therefore

$$t \ge \gamma + T(Ig - cf) + (\tau - \ell) - hX > 0.$$

The same witness satisfies a coherent zero factor of N. This establishes both directions without an unproved enlargement of a Pell index.

Exact degree. Before absorption, the eight factors have exact degrees

$$119, 119, 119, 119, 107, 107, 101, 97,$$

with nonzero leading factors, so the lean norm has degree 888. Writing N_{lean} for that unabsorbed norm, the exact identity

$$N = Y^{16} N_{\text{lean}}\left(t + \frac{k - R - 1}{Y}\right), \qquad s_{\text{old}} = s + 3q,$$

adds 64 degrees. The rational shift has degree 0, below t's degree 1, and cannot alter the leading form. The transport factor adds 2, giving 954. The full proof supplies every leading factor and a nonvanishing certificate uniform in every authentic coefficient tuple.

Twelve-positive companion. Retain positive h, use $Y = sq^3$, set $S = H$, $L = 1 + HIT$, $W = b_0 + H(t + b + \ell)$ and $N_k = k - hE - R$. Then $N_k N_t(1 + N) - 1 = 0$ has exact degree $888 + 7 + 2 = 897$ and costs 108 operations. Completeness uses the preceding formula for t without the term hX, and soundness uses the inherited common-unit signed theorem.

Validation and limits. Independent checks multiply the formal radicals, replay complete univariate polynomials over three prime fields using the actual fixed coefficient tuple, verify all coefficient payload hashes, and stress the no-index-equation bounds at $k = 2$ and arbitrarily larger X. A separate reviewer reproduced the analytic leading coefficient at nine points over three primes. Finite checks corroborate the identities and transcription; they do not replace the inherited unbounded compiler proof. The complete proof, its pinned source texts, source code, inert circuits, and replay receipts accompany this section.

6.1 Detailed recovery and degree arguments

This subsection supplies the details behind the positive construction's new steps. The complete parallel proof text is also retained as `positive/Y_ABSORPTION_PROOF.md`, alongside the pinned inherited signed-compiler proofs.

The literal norm evaluator. The eight-factor expression is evaluated without radicals by the following integer-polynomial identities. These registers make the equation fully explicit even without expanding its exponentially many monomials:

$$\begin{aligned}
d_n &= dS^2A, & H_Q &= S^2Q, & H_P &= H_QDv^2,\\
H_U &= H_Qb^2, & H_e &= H_P + H_U - S^2, & j_0 &= 4H_PH_U,\\
T_r &= 2d(W^2 + L^2G), & K_n &= [d(W^2 - L^2G)]^2, & &\\
A_a &= 2(d_n + H_e), & c_c &= (d_n - H_e)^2 - j_0, & &\\
a_t &= T_r - A_a, & b_t &= c_c - K_n, & c_o &= 16d_nj_0,\\
v_b &= c_o - a_t(a_tT_r + 2b_t), & u_r &= b_t^2 - a_t^2K_n, & &\\
N &= K_nv_b^2 - u_r(T_rv_b - u_r). & & & &
\end{aligned}$$

This is the universal quadratic-remainder norm identity. No specialized Galois independence is needed. With $x_0 = d(W^2 + L^2G)$, one has $T_r = 2x_0$ and $K_n \equiv x_0^2 \pmod 4$, hence $N \equiv (u_r - x_0v_b)^2 \pmod 4$. The unit equation $N_t(1 + N) = 1$ therefore excludes $N = -2$ and forces $N = 0, N_t = 1$.

Bounds before any missing quotient is recovered. From $N_t = 1$, if $C \le 0$, then

$$N_t \le q - F - z(q - 1) = 1 - F - (z - 1)(q - 1) \le 0,$$

a contradiction. The whole authentic tuple supplies the remaining outer bounds already stated. In particular $c > 2Y > 96$, $I \ge c^2$ and $a + 2 < 2E$. For the positive real roots $g = \sqrt{G}$, $f = \sqrt{D}$, $\tau = \sqrt{A}$, $g < 2Ec$ and $f < 2EI$. Direct expansion gives

$$1 < A < \Delta c^2 \le \Delta I = \sqrt{Q},$$

since $\Delta c^2 - A > k^2[EY(2Y^2 - 1) + Y^4] - 1 > 0$. Moreover $c < I^2$, $R < E$ and $D < 4E^2I^2$ give

$$b + cf < 6E^3I^2, \qquad d > \tfrac{1}{2}E^4I^2, \qquad \frac{b + cf}{d} < \frac{12}{E}.$$

Using $I^2 - c^2 > I^2/2$, $Ig + cf < 2Ig$ and $ic > 96$,

$$Ig - cf = \frac{I^2 - c^2}{Ig + cf} > \frac{I}{8Ec} = \frac{ic}{8E} > \frac{12}{E}.$$

These estimates do not assume $k > E$ or the absent index quotient.

Coherent inner roots, including the exceptional cases. For a formal variable x, put

$$C_x = d(x^2 + A) - e, \quad \mathcal{F}(x) = [C_x^2 + DB_a^2 - 4d^2x^2A]^2 - 4C_x^2DB_a^2.$$

Factoring quadratics shows that every root has a coherent description $x = \tau' + y$, $(\tau')^2 = A$, $(f')^2 = D$ and $dy^2 = Q(vf' - b)^2 - 1$, including possibly complex y. For $Z_b = b + vf > 1$, one has $|y|^2 \le (QZ_b^2 + 1)/d$

and $|y| < Z_b + Z_b/d$ because the difference of squares is $(QZ_b^2 - d)/d^2 > 0$. With $T \geq 1$ and the preceding bound, $|y| < b + ITg$ and $|x| < \tau + b + ITg$. This justifies the negative-main-root exclusion in the preceding section. If G were nonsquare, actual quadratic conjugation would transfer a root of the rational polynomial $\mathcal{F}$ to that excluded negative-main-root value. Thus g is integral and $x = (W - Lg)/S$ is rational.

If $C_x = 0$ at $\mathcal{F}(x) = 0$, substitution gives

$$d^2A^2 - deA + Q^2v^2b^2D = 0.$$

Modulo Q, this forces $Q \mid A(A-1)$, impossible since $0 < A(A-1) < Q$. Similarly $\mathcal{F}(0) = 0$ forces $Q \mid (A-1)^2$. Thus x, C_x are nonzero. Put

$$U_x = C_x^2 + DB_a^2 - 4d^2x^2A, \qquad f' = -\frac{U_x}{2C_xB_a}, \quad \tau' = \frac{C_x + B_af'}{2dx}, \quad y = x - \tau'.$$

The denominators are nonzero, and the polynomial identity verifies $(f')^2 = D$, $(\tau')^2 = A$ and the auxiliary Pell equation. The rational square roots f', τ' are integers.

The inherited rational Pell saturation lemma states that $m \geq 2$, $V \in \mathbb{Z}$, $y \in \mathbb{Q}$ and $m^2V^2 - (m^2-1)y^2 = 1$ imply $y \in \mathbb{Z}$. To recall its proof, write $y = p/r$ in lowest terms; $r^2 \mid m^2 - 1$. The integer-coordinate Pell equation with positive nonsquare discriminant $(m^2-1)/r^2$ has solutions (m, r) and $(m|V| - |p|)$. If their indices are j, l, abscissa divisibility forces l an odd multiple of j: reduce l modulo $2j$, reflect to $[0, j]$, and use $\chi_{n+2j} \equiv -\chi_n \pmod{\chi_j}$ and strict growth. The ordinate addition law now gives $r \mid p$, hence $r = 1$. Apply this lemma with $m = \Delta I$ and $V = vf' - b$. Thus y, x are integers. Reducing $W = Lg + Sx$ modulo Y gives only $Y \mid H(k - R - 1)$ at this stage; no cancellation of H is assumed.

The order of index recovery. Write $A_0 = a + 2$. The main and strong Pell norms give positive indices p, m with $g = \chi_{A_0}(p)$, $c = \psi_{A_0}(p)$, $|f'| = \chi_{A_0}(m)$, $I = \psi_{A_0}(m)$. From $I = ic^2$, strong divisibility yields $p \mid m$; write $m = pj$. Composition modulo c gives $c \mid j$, hence $c \mid m$ and $m > 2p$. Since $c > 1$, $p \geq 2$ and recurrence growth gives $c > 2p$; also $c > 6R$. For the auxiliary equation, set $V = c(Tf' - 1) - R(f')^2$; classification at ΔI gives an odd index j_0. The odd-index polynomial obeys

$$\chi_s(2r+1) = sQ_r(s^2), \quad Q_r(1 - A_0^2) = (-1)^r\psi_{A_0}(2r+1), \quad Q_r(0) = (-1)^r(2r+1).$$

Modulo $|f'|$, $Q \equiv 1 - A_0^2$ and $V \equiv -c$; the pinned signed index-reduction theorem, using $2p < m$, gives $j_0 \equiv \pm p \pmod{m}$, hence modulo c. Modulo c, $Q \equiv 0$, $V \equiv -R$, so $j_0 \equiv \pm R \pmod{c}$. As $0 < p, R < c/2$, $p = R$. These inherited steps precede both the main congruence modulo H and the missing index equation.

The first norm is $\tau'^2 - k^2EY(EY+1) = 1$, with fundamental unit $(2EY+1, 2)$. Put $P_0 = 2EY + 1$; then $|\tau'| = \chi_{P_0}(n)$, $k = 2\psi_{P_0}(n)$ for $n \geq 1$. Since $P_0 > A_0$ and $c > k$, monotonicity gives $n < p = R$. Now $\mu = Y/\gcd(Y, H) = Y/\gcd(Y, 3) > R$, and the recovered divisibility says $k \equiv R + 1 \pmod{\mu}$. Recurrence reduction gives $k \equiv 2n \pmod{Y}$. Thus $2n - R - 1$ is divisible by μ and has absolute value less than R, forcing $2n = R + 1$. Reduction modulo E also gives $k \equiv 2n \pmod{E}$, so $h = (k - R - 1)/E$ is integral. It is positive: $R > 48$ gives $n \geq 25$ and $k > 2n$.

Dividing the carrier by Y, using $(k - R - 1)/Y = hX$, now gives $g \equiv b_0 \pmod{H}$. Since $n \geq 2$ implies $k \geq 4P_0 > X$, one has $c > X$ and $g > (a+1)c$, so $\gamma = (g - b_0)/H > 0$. The ordinary signed compiler is reached literally by $i_{\text{ordinary}} = \Delta i$ and $I_{\text{ordinary}} = \Delta I$:

$$I_{\text{ordinary}}^2 = \Delta((f')^2 - 1) = Q.$$

If $f' < 0$, change f', T simultaneously to their negatives. The retained ordinary signed theorem permits that signed auxiliary quotient and extracts a blank K215 halt. No stronger unquoted normalized theorem is needed.

A complete leading-form witness. For the unabsorbed twelve-positive lean norm define

$$q_0 = \beta J,\ X_0 = wq_0,\ Y_0 = sq_0^3,\ E_0 = X_0Y_0,\ k_0 = \eta + \zeta,$$
$$c_0 = k_0Y_0,\ I_0 = ic_0^2,\ \ell_0 = E_0c_0,\ H_0 = 4E_0,\ d_0 = E_0^4I_0^2,$$
$$R_0 = q_0^3(q_0 - F),\ B_0 = R_0E_0^2I_0^2,\ V_0 = I_0T\ell_0.$$

The degrees of $H, \ell, d, b, IT\ell$ are $6, 11, 46, 38, 23$. The positive-leading formal Laurent branches have

$$\tau - \ell = k/2 + \text{lower terms}, \qquad (b_0 + g)/H = c_0/2 + \text{terms below degree 5},$$

while $(b_0 - g)/H$ has degree -1 and $T(Ig - cf) = iTc_0/(2E_0) +$ terms below degree 1. These expansions concern degree, not rationality of specialized roots. For each sign triple put $Z = W_0 - Hb - H\varepsilon_a\tau - L_0\varepsilon_g g$. The corresponding factor equals

$$2dHbZ + dZ^2 - H^2b^2 - H^2QDv^2 + H^2 + \varepsilon_f 2H^2Qbvf.$$

For $\varepsilon_f \neq \varepsilon_g$, four factors have degree 119 and leader $-\varepsilon_g 4H_0^2d_0B_0V_0$. For matched signs, the degree-119 terms cancel. The identity

$$dI^2G - QDc^2 = Q(I^2 - c^2) - I^2G$$

handles the leading-square cancellation; the remaining cross term has degree at most 92 and H^2b^2 has degree 88. The other four leading terms are

$$\begin{aligned}
&\varepsilon_a = -1,\ \varepsilon_g = \pm 1: && 107, && 4H_0^2d_0B_0\ell_0,\\
&\varepsilon_a = +1,\ \varepsilon_g = -1: && 101, && H_0^2d_0B_0c_0,\\
&\varepsilon_a = +1,\ \varepsilon_g = +1: && 97, && H_0^2d_0B_0(2t - k_0 - iTc_0/E_0).
\end{aligned}$$

Their product is the nonzero leading form

$$2^{12}H_0^{16}d_0^8B_0^8V_0^4\ell_0^2c_0(2t - k_0 - iTc_0/E_0).$$

It is nonzero because t is independent. Hence the norm degree is exactly $4 \cdot 119 + 2 \cdot 107 + 101 + 97 = 888$. For eleven variables, the exact identity $N = Y^{16}N_{\text{lean}}(t + (k - R - 1)/Y)$ and $s_{\text{old}} = s + 3q$ adds 64 degrees. The rational shift has degree 0, below t's degree 1; the carrier degree is at most 16, so denominators clear. The affine s substitution has invertible top linear part $s + 3\beta J$, preserving nonvanishing. The transport factor has nonzero degree-2 leader $w(q_0 - F - \alpha) - zq_0$. This proves degrees 952 and 954 for the norm and product, for every authentic $\beta > 0$, not merely sampled coefficients.

7 Reproduction and verification boundary

7.1 One command from a clean extraction

From any directory run Python on the extracted release's `verify_release.py`. It requires Python 3.10 or later, SymPy, and NumPy. The command first rejects optimized Python execution (`-O` and `-OO`), verifies the exact manifest file set and every SHA-256, and checks all retained source Git blob identities. It then creates an isolated temporary copy and executes only the newly authored checks there. Any check failure causes a nonzero exit. It does not rewrite the sealed release or import an upstream program. No network or proof-assistant install is required for the finite replay.

The archive contains the editable LaTeX source and rendered PDF; all exact JSON arithmetic circuits and degree ledgers; raw and compressed native coefficients; compressed signed derived coefficients with precise encodings; all small isolation certificates; the newly authored builders and verifiers; source proofs,

exact pins, and inherited licenses/notices. For a normal TeX Live or MiKTeX installation, run `pdflatex report.tex` twice from `report/`, or use `latexmk -pdf report.tex`. The supplied shell helper is a Linux fallback for the research environment; it uses its existing TeX installation without shell escape. The environment-specific font/format cache is not shipped.

The archive-level manifest is integrity metadata, not an external digital signature. Its own file cannot contain its own hash. The outer ZIP hash should be recorded by the recipient if a trusted transport digest is desired.

7.2 Checks actually supported by the scripts

1. Reconstruct the complete native tuple from the literal tile list and verify its bounded decoder, synchronization, rank-controller conditions, and every payload hash.
2. Expand all 22 shifted parent residuals in a symbolic integer polynomial ring, deriving the 136 weighted records and their actual coefficient bounds.
3. Exhaustively check each source-record/monomial/target collision condition, including padding translations and coefficient signs.
4. Prove the shared source circuit equals the explicit source in $\mathbb{Z}[B, \beta, K_0, M_C, M_F]$, with 159 port-monomial terms. This is exact symbolic identity checking, not random polynomial testing.
5. Replay the full nine-variable source and finalizer with the actual tuple, its separately translated domains, the Pell-majorant identities and norm obstruction/completeness fixtures. A separately authored audit checks the same full circuits and leading coefficient.
6. Replay every retained ten-integer source and tail with signed test inputs; check its polynomial identities and nonzero top coefficients over three finite fields. Circuit degree propagation gives the matching upper bound, so a nonzero diagonal coefficient proves exact total degree.
7. Multiply the positive construction's formal radicals independently, replay full univariate polynomials over three finite fields using the actual coefficient tuple, check the unit guard, and test its pre-index bounds. A separate leading-form audit compares the analytic leader with the literal circuit.
8. Independently traverse the important pinned 38/44 circuits for all live binary gates and compare the cost–degree frontier against the correct optional variants, not only headline forms.

Numerical fixtures are finite diagnostics. Exact symbolic identities and nonzero modular leading coefficients have the stated finite mathematical force; neither proves an unbounded compiler theorem by itself. The newly written proofs, composed with the stated inherited theorems, provide the ordinary mathematical consistency equivalence formalizable in ZFC.

7.3 Formalization status at the later pin

The postscript separately records `2d7766807b92f6d505649885d472e20cc0236b4f`, later than the construction snapshot. Its exact status and compiler documents report an executable supplied-data Jones compiler, sparse coefficient reader, explicit index construction, the full primitive-recursive graph compiler, and an externally checked nine-variable ZFC endpoint whose coefficients remain existence-based. Extraction of the actual refutation program to `ScalarPRCode.Realizes haltingTest` remains open. The ordinary P67 compiler proof and a checked object-level ZFC derivation remain separate open integration boundaries.

We have not rerun those Lean builds here and do not claim that the new literal equations in this archive have been kernel checked. The new sparse quadratic interface is an attractive integration target for that executable compiler; the later progress does not retroactively certify these circuits.

8 Further research questions

1. Can support-local collision avoidance and target placement be optimized jointly with addition-chain cost, rather than minimizing only the maximum position or target span? A proof-producing search can emit the same small isolation certificate independently of its heuristic.
2. Can a third-order or asymmetrically scaled polynomial Pell majorant improve the exact nine-variable degree after all cancellation terms are charged? Its strict positivity must hold before squarehood is known.
3. Is there a safe analogue of conditional separation for the unrestricted-integer three-radical norm? The negative-radicand and zero-scale branches make a naive removal of its separators invalid.
4. Can an additional quotient be absorbed without enlarging the witness count, while preserving both directions and the exact positive domains? The CRT change here is a bijection, not a claim of cofinal Pell witnesses.
5. Can package 40's 457-selector factorized-receiver candidate be made a complete compiler? Its empty cells may retain payload bits; an occupancy-first interface proof is needed before its coefficients may replace a complete tuple.
6. Can the new sparse quadratic interface be connected directly to the post-pin executable supplied-data Jones compiler, avoiding the quartic sum-of-squares detour? Concrete refutation-program extraction and the object-level ZFC derivation are distinct remaining tasks.

These are bounded next directions, not completed improvements. The present release already specifies the exact equations and their reproducible finite certificates without depending on any future success.

References and provenance

1. ProveIt construction snapshot: the pinned construction collection at commit 124c07c0b77a. Full file URLs and Git blob hashes are in `review/SOURCE_PINS.json`.
2. J. P. Jones, Universal Diophantine equation, *Journal of Symbolic Logic* 47 (1982), 549–571. https://doi.org/10.2307/2273588. The exact inherited ratio interfaces are preserved in `inherited/nine/NineRatio.lean`; no new compilation is claimed here.
3. Zhi-Wei Sun, Further results on Hilbert's tenth problem, https://arxiv.org/abs/1704.03504, version 7 (2021), and the later arithmetic bridge cited by the pinned package 31, https://arxiv.org/abs/2609.01594, version 2 (2026). The exact mathematical dependency is the retained `inherited/sun/SIGNED10_PROOF.md`, Sections 4–8.
4. Native coefficient/compiler proofs: package 44 `native/PROOF.md` and package 39 `optional_compact/coefficients/PROOF.md`. The signed ordinary compiler boundary is preserved as `positive/upstream/ORDINARY_BLANK_PROOF.md`.
5. Post-pin status snapshot: the pinned status documents at commit 2d7766807b92. The four exact status/compiler documents are included with their own pins.